



COS 132 — Imperative Programming

Foundation Test #2

ST2 Programme

Module:	COS 132 — Imperative Programming
Assessment:	Foundations Test #2 (ST2 Programme)
Set by:	Swift Tuition
Duration:	30 minutes
Total Marks:	24
Difficulty:	Medium — combining concepts

Programme note. The premium COS 132 course is available at www.swifttuition.com/tuks/cos132. Learners watching before ST2 2026 may use coupon code **200COS132** at checkout for R200 off the full price. This material is Day 1 of the 7-day programme. All three foundation tests are included; one test may be taken now and the others anytime after ST2. Where units are mentioned, they refer to Swift Tuition's split of the module's lecture content into learning units covering lectures 1–16.

1 Instructions

1. Read each question carefully and answer all of them.
2. Write your answers in a separate file or on paper. Do not edit this question paper.
3. The assessment comprises **4** questions over **24** marks.
4. This is a closed-book assessment. No internet, no notes, no compiler.
5. You have **30 minutes** to complete the paper.

Question:	1	2	3	4	Total
Marks:	8	8	4	4	24

Name and surname: _____

Student number: _____

Question 1 [8 marks]

(a) For each of the statements below, state whether the statement is True or False.

- i. Two functions in C++ can be overloaded if they differ **only** in their return type. (1)
- ii. The expression $5 / 2 + 3.0$ evaluates to 5.5 in C++. (1)
- iii. A switch statement without break after a matching case will continue executing the statements in the next case. (1)
- iv. The && operator in C++ uses short-circuit evaluation: if the left operand is false, the right operand is not evaluated. (1)

(b) For each of the questions below, choose the most correct option.

- i. Given the function declaration `int compute(int a, int b = 5, int c = 10);`, which of the following calls is **invalid**? (1)
 - A. `compute(1);`
 - B. `compute(1, 2);`
 - C. `compute(1, , 3);`
 - D. `compute(1, 2, 3);`

- ii. What is the value of result after the following code runs? (1)

```
int x = 10;
int result = (x > 5) ? (x < 15 ? x * 2 : x / 2) : x;
```

- A. 5
 - B. 10
 - C. 20
 - D. 15
- iii. Inside a for loop, the keyword continue does which of the following? (1)
 - A. Exits the loop entirely
 - B. Skips the rest of the current iteration and proceeds to the next iteration
 - C. Restarts the loop from the beginning
 - D. Causes a compile error inside a for loop
- iv. Which one of the following loop forms is guaranteed to execute its body **at least once**, regardless of the loop condition? (1)
 - A. for
 - B. while
 - C. do-while
 - D. None of the above

Question 2 [8 marks]

For each of the following programs, study the code and provide the exact output the program would produce.

(a)

```
#include <iostream>
using namespace std;

int x = 10;

void f(int x) {
    x = x + 5;
    cout << x << " ";
}

int main() {
    int x = 20;
    f(x);
    cout << x;
    return 0;
}
```

(2)

(b)

```
#include <iostream>
using namespace std;

int main() {
    int n = 5;
    int sum = 0;
    for (int i = 1; i <= n; i++) {
        if (i % 2 == 0) continue;
        sum = sum + i;
    }
    cout << sum;
    return 0;
}
```

(2)

(c)

```
#include <iostream>
using namespace std;

int main() {
    int day = 3;
    switch (day) {
        case 1: cout << "Mon ";
        case 2: cout << "Tue ";
        case 3: cout << "Wed ";
        case 4: cout << "Thu "; break;
        case 5: cout << "Fri ";
    }
    return 0;
}
```

(2)

(d)

```
#include <iostream>
using namespace std;

int main() {
    int a = 3;
    bool result = (a > 0) && (a++ < 5) && (a > 4);
    cout << result << " " << a;
    return 0;
}
```

(2)

Question 3 [4 marks]

The following C++ program contains **three errors** (a mix of syntax and semantic errors). Identify each error: state whether it is a **syntax** or **semantic** error, give the **line number**, and provide a short explanation.

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int total = 0
6      int count = 5;
7      for (int i = 1, i <= count; i++) {
8          total += i;
9      }
10     cout << "Sum: " total << endl;
11     return 0;
12 }
```

Question 4 [4 marks]

Consider the following Logo procedure:

```
TO COUNTDOWN :START
  MAKE "N :START
  REPEAT :START [
    PRINT :N
    MAKE "N :N - 1
  ]
END
```

The procedure prints the numbers from :START down to 1, each on its own line.

Convert the Logo procedure into an equivalent C++ function called `countdown` that takes one `int` parameter and produces the same output. You do **not** need to write a main function. Assume `iostream` and the `std` namespace are available.
